



Heap



- A **binary heap** is a simple data structure that efficiently supports the priority queue operations.
- A **binary heap** (or simply **heap**) is a complete binary tree in which the key value stored at any vertex is either less than or greater than the key values of its children.
- Types – Two
 - **Max-heap** or **descending heap**: Value present at any node is greater than all its children.
 - **Min-heap** or **ascending heap**: Value present at any node is less than all its children.

Heap

Unit IV. 1

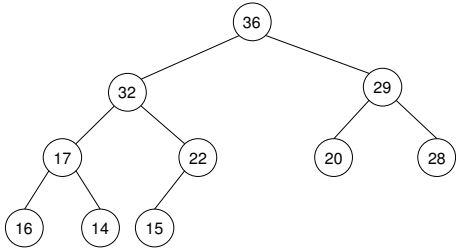
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1



Example: Max-Heap





Elements in Corresponding Array Representation:
36, 32, 29, 17, 22, 20, 28, 16, 14, 15

Heap

Unit IV. 2

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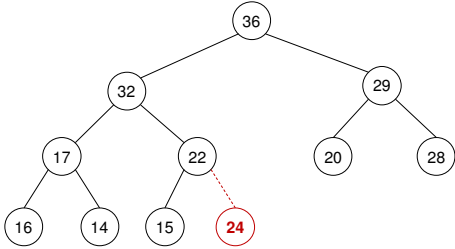
2



Insertion of node in Heap



■ Insertion of 24



Elements in Corresponding Array Representation:
36, 32, 29, 17, 22, 20, 28, 16, 14, 15, 24

Heap

Unit IV. 3

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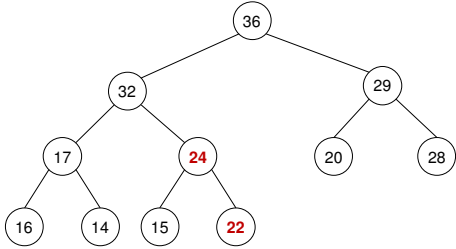
3



Insertion of node in Heap



■ Insertion of 24



Elements in Corresponding Array Representation:
36, 32, 29, 17, 24, 20, 28, 16, 14, 15, 22

Heap

Unit IV. 4

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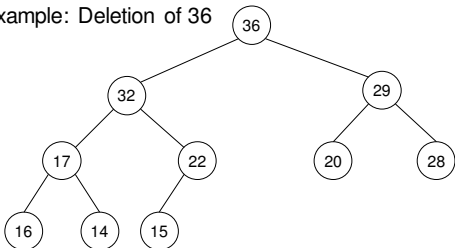
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Deletion of a node from Heap



- The node to be deleted from Heap is always of highest priority, i.e., **root node**.
- Example: Deletion of 36



Elements in Corresponding Array Representation:
36, 32, 29, 17, 22, 20, 28, 16, 14, 15

Heap

Unit IV. 5

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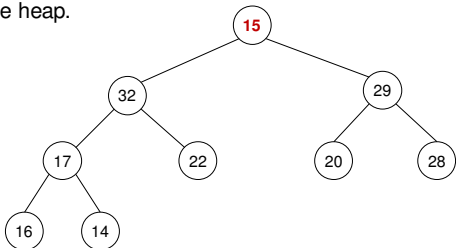
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Deletion of a node from Heap



- Copy the value at last index (i.e., 15 in our example), at the index 1, decrease the array index by one and restore the heap.



Elements in Corresponding Array Representation:
15, 32, 29, 17, 22, 20, 28, 16, 14

Heap

Unit IV. 6

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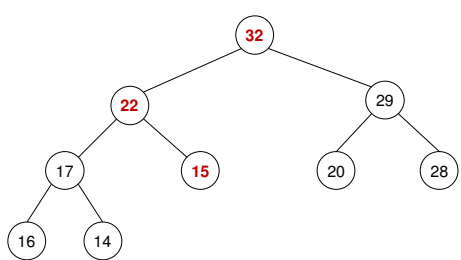
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Deletion of a node from Heap



■ Restore the heap



Elements in Corresponding Array Representation:
32, 22, 29, 17, 15, 20, 28, 16, 14

Heap

Unit IV. 7

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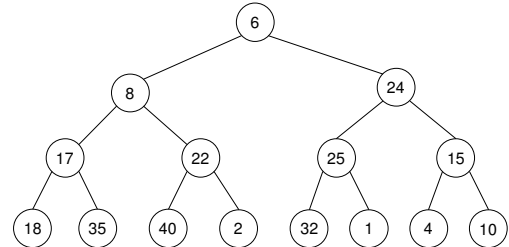
Construction of Heap



■ Example: Construct Max-heap from

□ 6, 8, 24, 17, 22, 25, 15, 18, 35, 40, 2, 32, 1, 4, 10

Tree Construction from the elements in Array



Heap

Unit IV. 8

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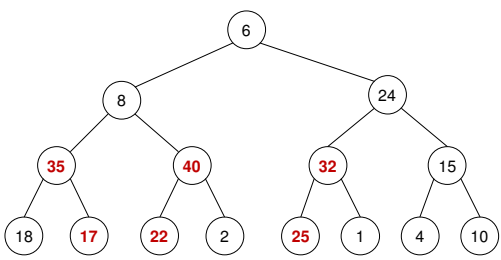
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Construction of Heap (Contd...)



Step 1



Elements in Corresponding Array:
6, 8, 24, 35, 40, 32, 15, 18, 17, 22, 2, 25, 1, 4, 10

Heap

Unit IV. 9

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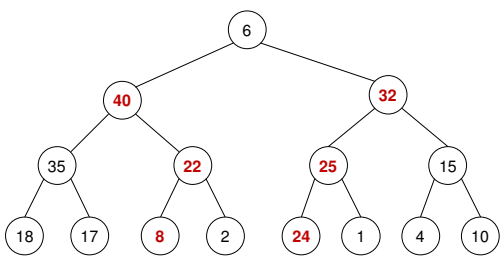
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Construction of Heap (Contd...)



Step 2



Elements in Corresponding Array:
6, 40, 32, 35, 22, 25, 15, 18, 17, 8, 2, 24, 1, 4, 10

Heap

Unit IV. 10

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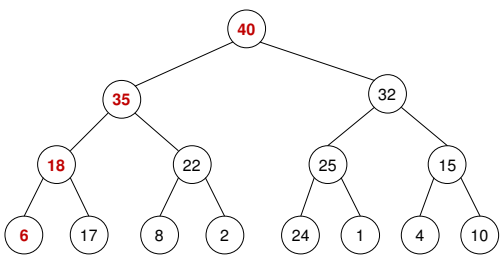
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Construction of Heap (Contd...)



Step 3



Elements in Corresponding Array:
40, 35, 32, 18, 22, 25, 15, 6, 17, 8, 2, 24, 1, 4, 10

Heap

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Exercise



■ Construct the min-heap from the given values

□ 6, 8, 24, 17, 22, 25, 15, 18, 35, 40, 2, 32, 1, 4, 10

Heap

Unit IV. 12

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